



Azure Cloud

Orchestration & Monitoring

PL_Master

runs every step in
order

QualityCheck

Azure Function

UpdateStats

Azure Function

Logic App

success / failure
email

Azure Monitor

failed-run alert

Key Vault

email URL secret



Azure Data Factory

Data Sources

1



KAUST Repository
CSV • 2023 , API



KSU Open Data Portal
JSON • 2023–2025



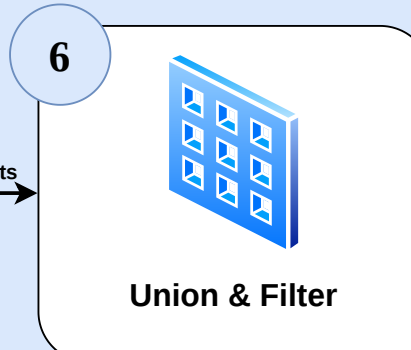
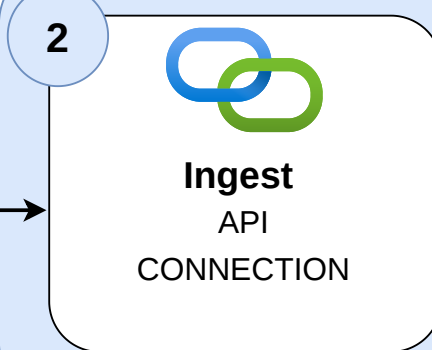
KFUPM Pure
MODS XML • OAI-PMH



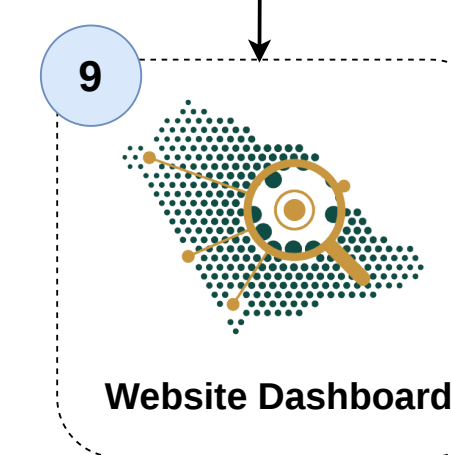
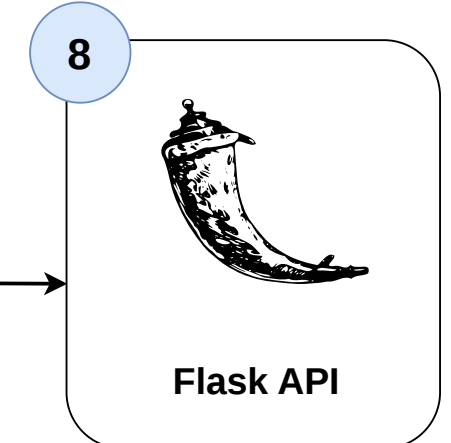
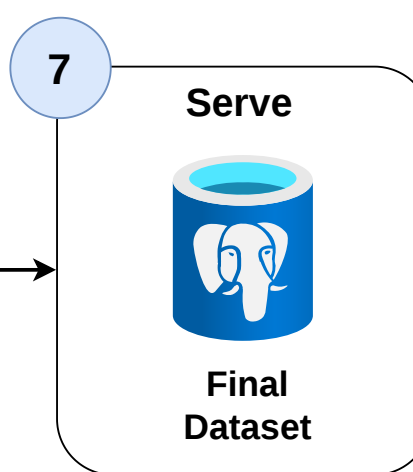
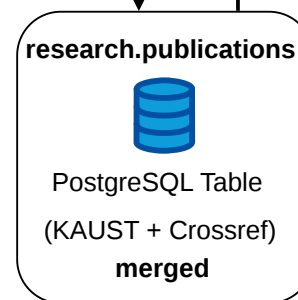
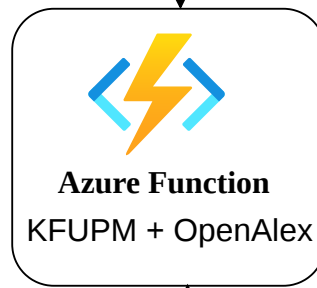
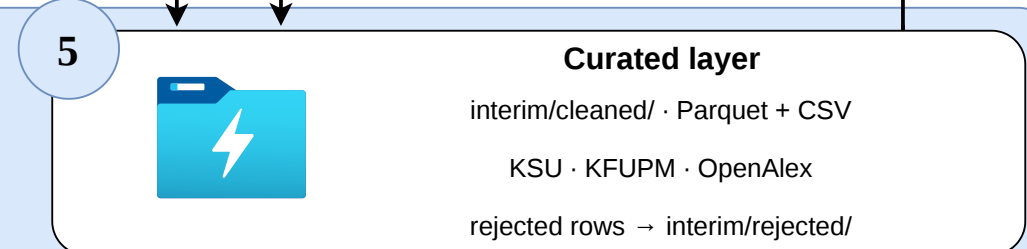
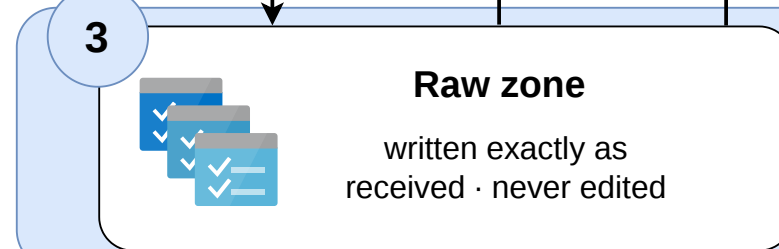
Crossref REST API
JSON • 2024–2026



OpenAlex
JSON • REST API
KAU • KKU • PSAU



Data Lake Storage Gen2



Data Governance



Data quality

Validation rules V1–V9
Rejected rows kept with reason
Checked before publishing
Stops if data drops 20%



Lineage

Raw → Cleaned → PostgreSQL
Raw files never edited
Every row keeps its source
Enrichment has evidence file



Access

Roles: Owner, Contributor, Reader
Functions use managed identity
No stored keys
Readers: SELECT only



Change & audit

Pipelines versioned in GitHub
Run history in Data Factory
Quality checks logged
Stats saved per run